

February 23, 2026

ECEN-607: Advanced Analog IC Design

Assignment #3

Due: March 2, 2026

Homework will not be received after due.

Upload your solutions through Canvas

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Design at transistor level with a low-frequency gain equivalent to a three-stage **single ended** amplifier; use the following specifications:

$V_{DD} = 1.8 \text{ V}$

$V_{SS} = 0 \text{ V}$

Loop Gain $> 60 \text{ dB}$ at low frequency

Gain $> 40 \text{ dB}$ at 20 MHz

Phase margin $> 45^\circ$

Load Capacitor = 5 pF

Minimize power consumption

Remark: You do not have to deal with the CMFB in this assignment.

- i) Select the topology; it could be a 3-stage cascade of amplifiers or a two stage with cascode output stage or even folded cascode stage. Find the transistor dimensions and value of the bias currents.

Simulate the amplifier and plot both magnitude and phase response. Be sure the phase margin is over 45° . Include screen shots.

- ii) Implement a (closed-loop) capacitive amplifier with an inverting gain of 6 dB , $\beta=1/3$. For your simulations you may need to connect a large resistor ($>1 \text{ Mohm}$) in parallel with the feedback capacitor; why?
- Simulate the amplifier and report both magnitude and phase response. Include screen shots and identify the unity gain frequency and phase margin.
 - Plot the step response for a differential input step of 0 to 100 mV variation. Report the 1% settling time and value of the settling error when reaching steady state. Ideal output voltage variation should be -100 mV .
 - Comment on the results.

- iii) Repeat ii) for the case for closed-loop gain=12dB; evidently the feedback factor is different now. For the transient simulation, reduce the amplitude of the input step to 50mV.

Please discuss your results.